# *Uptake in supraclavicular area fat ("USA-Fat"): description on 18F-FDG PET/CT.*

The supraclavicular region is a common site for lymph node metastases. A commonly reported type of nonmalignant (18)F-FDG uptake on PET imaging in the supraclavicular region is "muscle uptake" purportedly due to muscle contraction in tense patients during the (18)F-FDG uptake phase. PET/CT offers the unique opportunity to correlate PET findings with CT anatomy in the supraclavicular region. METHODS: Images from the first 359 consecutive clinical whole-body studies (in 347 patients) using (18)F-FDG and a PET/CT scanner (with CT attenuation correction and ordered-subsets expectation maximization [OSEM] reconstruction) were retrospectively reviewed. The supraclavicular region was evaluated for the presence of abnormal uptake on PET images, and the corresponding CT findings were assessed. Three distinct patterns of abnormal (18)F-FDG uptake were noted: pattern A (uptake localizing to supraclavicular area fat [USA-fat], i.e., without corresponding lymph node or muscle uptake on CT), pattern B (uptake localizing to muscle on CT), and pattern C (uptake localizing to lymph nodes or soft-tissue masses on CT). RESULTS: Forty-nine patients (14.1%) (32 female, 17 male; mean age, 51.4 +/- 15.6 y; age range, 12-77 y) showed abnormal (18)F-FDG uptake in the supraclavicular region. Twenty patients (5.8%) had muscle uptake (group B); 15 (4.3%) had definite abnormal lymph nodes (group C). However, 14 patients (4.0%) had USA-fat (group A) and foci of very low Hounsfield units on CT. These foci were also present on (68)Ge attenuation-corrected images (when obtained) and non-attenuation-corrected images. Uptake in USA-fat was typically bilateral and symmetric, intense, more often multifocal than linear, and located in fat on PET/CT. Age was not significantly different for group C versus the 2 other groups. Intensity; mean standardized uptake value, lean (SUV(L MEAN)); or maximum standardized uptake value, lean (SUV(L MAX)), did not allow differentiation between patterns A and C ($P > 0.05$). Standardized uptake values (SUV(L MAX), 3.1; SUV(L MEAN), 2.1) were significantly lower in group B than in the 2 other groups ($P < 0.005$). CONCLUSION: So-called muscle uptake in the supraclavicular region may be caused in a significant proportion of cases by an unrelated process we call the USA-fat finding, with (18)F-FDG uptake in tissues of low-Hounsfield (fat) density. This finding most likely reflects an underlying nonpathologic process that we hypothesize to be in foci of brown fat. This intense supraclavicular uptake should be recognized and should not be misinterpreted as a malignant metastatic process or as muscle uptake.